DEMO101 Lecture 1

October 13, 2026 | Simple linear regression and least squares
Synthetic material for the UofT AI workshop

The model

We describe a response y using one predictor x:

```
y = beta0 + beta1 x + error
```

- beta0 is the intercept: expected y when x = 0.
- beta1 is the slope: expected change in y for a one-unit increase in x.
- The error term represents variation the line does not explain.

Least squares

Choose the line that makes the sum of squared vertical distances as small as possible:

```
minimize sum_i (y_i - b0 - b1 x_i)^2

b1 = Sxy / Sxx
Sxy = sum_i (x_i - x_mean)(y_i - y_mean)
Sxx = sum_i (x_i - x_mean)^2
b0 = y_mean - b1 x_mean
```

The fitted line passes through (x_mean, y_mean).

Interpreting the fit

- Slope: for each extra unit of x, y changes by b1 on average.
- Intercept: meaningful only if x = 0 is inside or near the observed x range.
- Do not predict far outside the observed x values; that is extrapolation.